

Physical Chemistry (I)

Lecture 06

First Law of Thermodynamics

Enthalpy



In the Last Class

- Real gas
 - van der Waals equation
 - van der Waals equation and virial expansion
 - Beyond van der Waals
- The first law
 - The first law of thermodynamics
 - Work

van der Waals Equation

- **Repulsion:** decrease of the total volume

$$V \rightarrow V - nb$$

$$p = \frac{nRT}{V - nb}$$

- **Attraction:** decrease of the pressure

$p \sim$ (frequency of collisions) $\times$ (force of each particle)

decreases as $n/V \uparrow$

decreases as $n/V \uparrow$

$$p = \frac{nRT}{V - nb} - a \frac{n^2}{V^2}$$

van der Waals and Virial

- Virial expansion

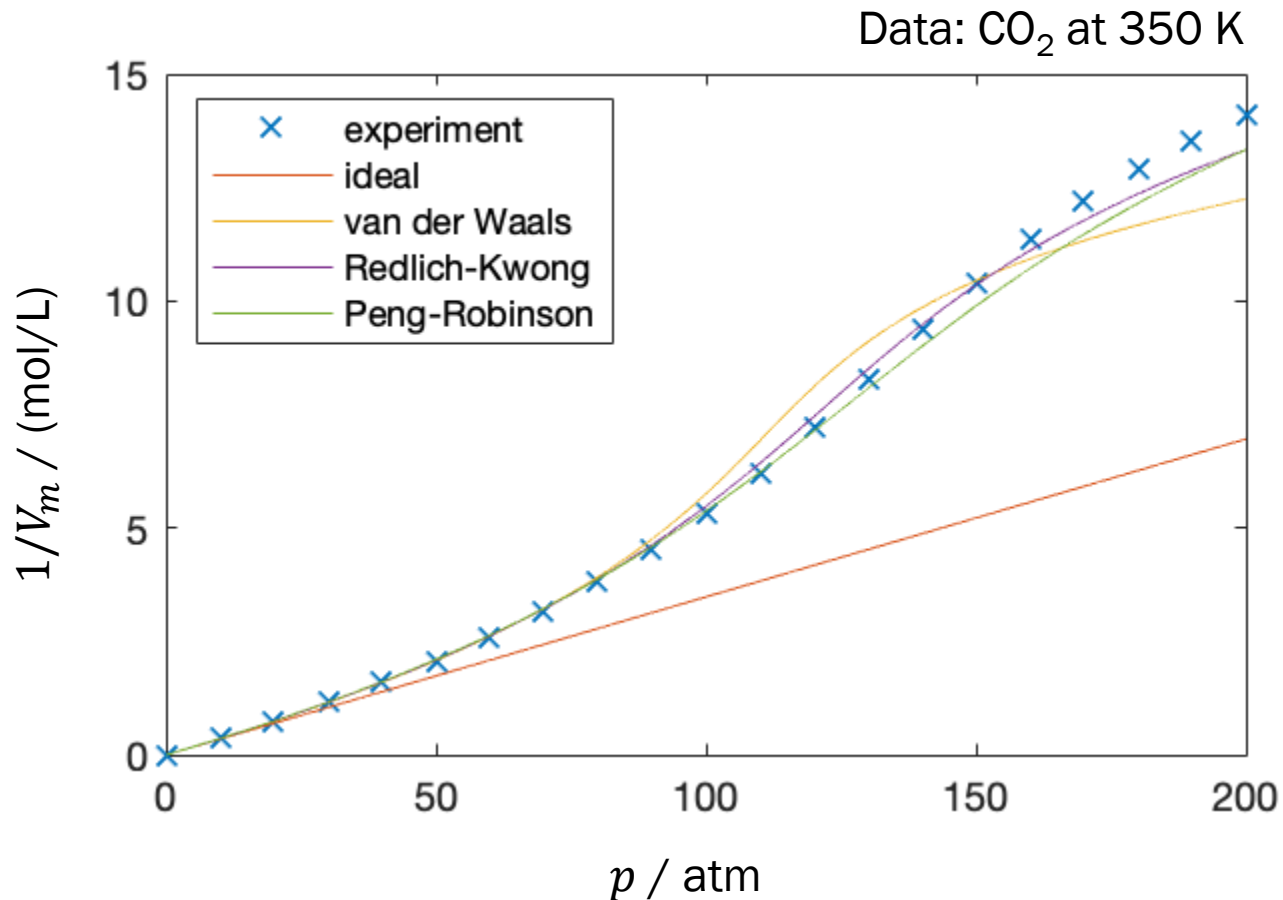
$$pV_m = RT \left(1 + \frac{B}{V_m} + \frac{C}{V_m^2} + \dots \right)$$

- van der Waals equation

$$p = \frac{RT}{V_m - b} - \frac{a}{V_m^2}$$

$$pV_m = RT \left[1 + \left(b - \frac{a}{RT} \right) \frac{1}{V_m} + \frac{b^2}{V_m^2} + \dots \right]$$

Beyond van der Waals



Last class

First Law of Thermodynamics

“The internal energy of an isolated system is constant.”

- Heat and work are equivalent ways of changing the internal energy of a system

$$\Delta U = q + w$$

$$dU = \delta q + \delta w$$

↑ ↑ ↑
small change of U small q small w



Last class

Calculation of Expansion Work

$$\delta w = -p_{\text{ex}} dV$$

- Total work for a change from state i to state f

$$w = \int_{\text{state } i}^{\text{state } f} \delta w = \int_{\text{state } i}^{\text{state } f} (-p_{\text{ex}} dV) = - \int_{V_i}^{V_f} p_{\text{ex}} dV$$

1. Expansion against constant external pressure
2. Reversible expansion



Last class

Internal Energy

$$dU = \delta q + \delta w \text{ (1st law)}$$

For $\delta w_{\text{add}} = 0$,

$$dU = \delta q - p_{\text{ex}}dV$$

How to determine δq ?

Heat

At constant volume,

$$\delta q_V \propto dT$$

$$\delta q_V = C_V dT$$

At constant pressure,

$$\delta q_p \propto dT$$

$$\delta q_p = C_p dT$$

Heat Capacity

- Molar heat capacity $C_m = C/n$
- Specific heat capacity $C_s = C/m$
- Heat capacity depends on the constraint
 - C_V : heat capacity at constant volume ($C_{V,m}$, $C_{V,s}$)
 - C_p : heat capacity at constant pressure ($C_{p,m}$, $C_{p,s}$)

Calculation of Heat (Const V)

$$\delta q_V = C_V dT$$

- Total heat for a change from **state i** to **state f**

$$q_V = \int_{\text{state } i}^{\text{state } f} \delta q_V = \int_{\text{state } i}^{\text{state } f} (C_V dT) = \int_{T_i}^{T_f} C_V dT$$

Internal Energy (Const V)

$$dU = \delta q - p_{\text{ex}}dV$$

At constant volume, $dV = 0$ and $\delta q = C_V dT$

$$dU = C_V dT$$

The change in internal energy is

$$\Delta U = \int_{\text{state } i}^{\text{state } f} dU = \int_{\text{state } i}^{\text{state } f} (C_V dT) = \int_{T_i}^{T_f} C_V dT$$

At Constant V

$$\Delta U = q_V = \int_{T_i}^{T_f} C_V dT$$

If C_V is constant,

$$\Delta U = C_V \int_{T_i}^{T_f} dT = C_V(T_f - T_i) = C_V \Delta T$$

Not in Atkins

(Monatomic) Perfect Gas

$$U = \frac{3}{2}nRT$$

$$dU = \frac{3}{2}nRdT$$

$$dU = \delta q - p_{\text{ex}}dV$$

$$\frac{3}{2}nRdT = \delta q - p_{\text{ex}}dV$$

Not in Atkins

Perfect Gas at Constant V

$$\frac{3}{2}nRdT = \delta q - p_{\text{ex}}dV$$

At constant volume, $dV = 0$ and $\delta q = C_V dT$

$$\frac{3}{2}nRdT = C_V dT$$

Hence,

$$C_V = \frac{3}{2}nR$$

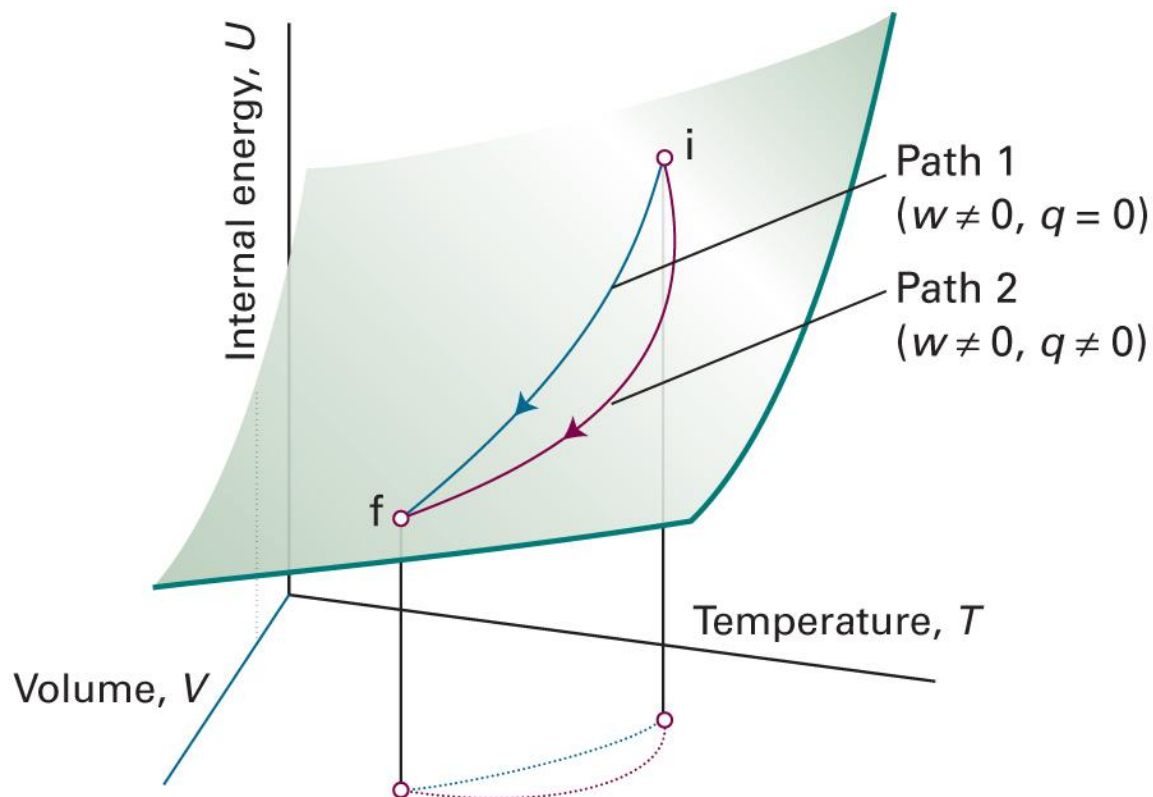
Now, get ready for some math...



State and Path Functions

- **State function:** a property that depends only on the current state of a system and is independent of its history
e.g. internal energy, pressure, volume, temperature
- **Path function:** a property with a value that depends on the path between two states
e.g. heat, work

State and Path Functions



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 2D.1)

Related Mathematics

- **Exact differential:** an infinitesimal quantity that, when integrated, gives a result that is independent of the path between the initial and final states.

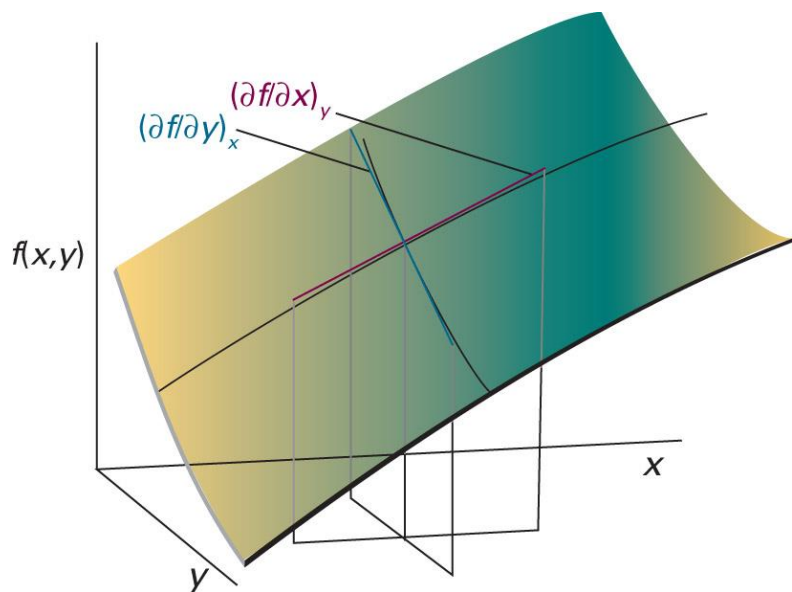
$$\Delta U = \int_{\text{state } i}^{\text{state } f} dU$$

- **Inexact differential:** an infinitesimal quantity that, when integrated, gives a result that depends on the path between the initial and final states.

$$q = \int_{\text{state } i, \text{ path}}^{\text{state } f} \delta q$$

Partial Derivatives

“the slope of the function with respect to one of the variables, all the other variables being held constant”



$$\left(\frac{\partial f}{\partial x}\right)_y = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x, y) - f(x, y)}{\Delta x}$$

$$\left(\frac{\partial f}{\partial y}\right)_x = \lim_{\Delta y \rightarrow 0} \frac{f(x, y + \Delta y) - f(x, y)}{\Delta y}$$

Differential

If f is a function of x and y , the differential df is

$$df = \left(\frac{\partial f}{\partial x} \right)_y dx + \left(\frac{\partial f}{\partial y} \right)_x dy$$

when df is an **exact differential**.

State Variables

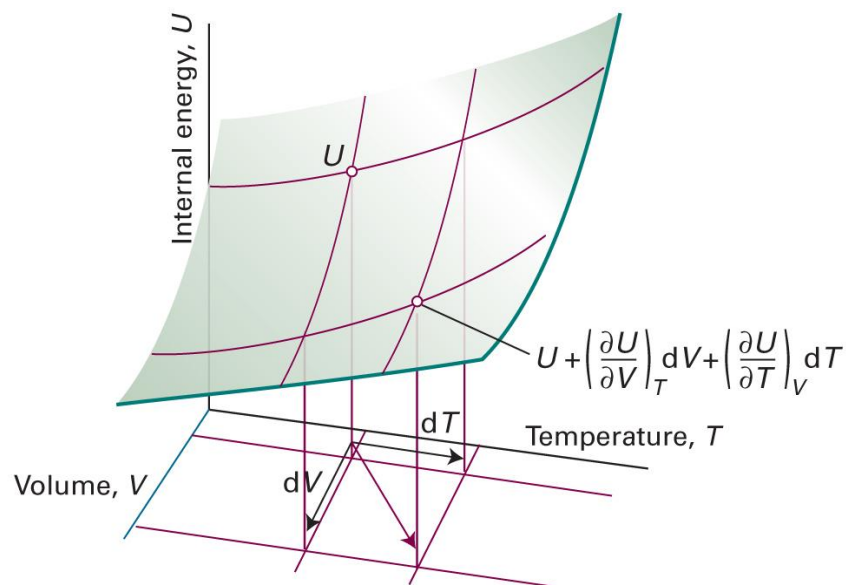
- A state is defined by p, V, n, T
- Typically n is fixed
- Equation of state: p is constrained by V and T
- We can just use V and T to define a state

$$U = U(V, T)$$

Changes in Internal Energy

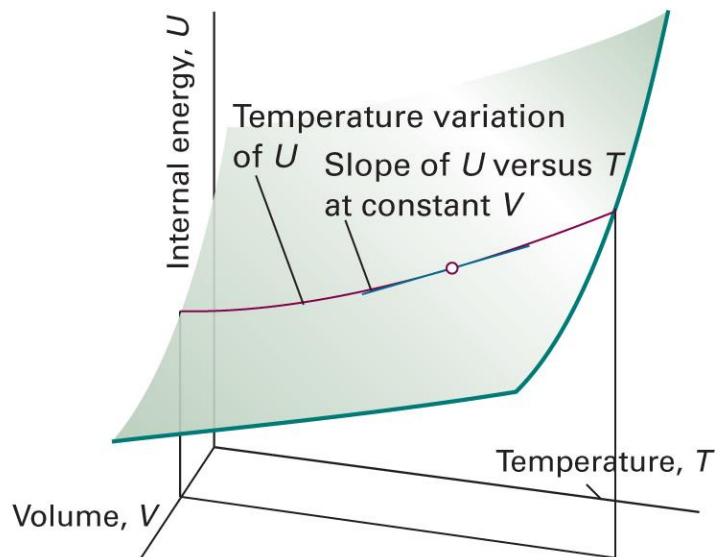
For $U = U(V, T)$,

$$dU = \left(\frac{\partial U}{\partial V} \right)_T dV + \left(\frac{\partial U}{\partial T} \right)_V dT$$



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 2D.2)

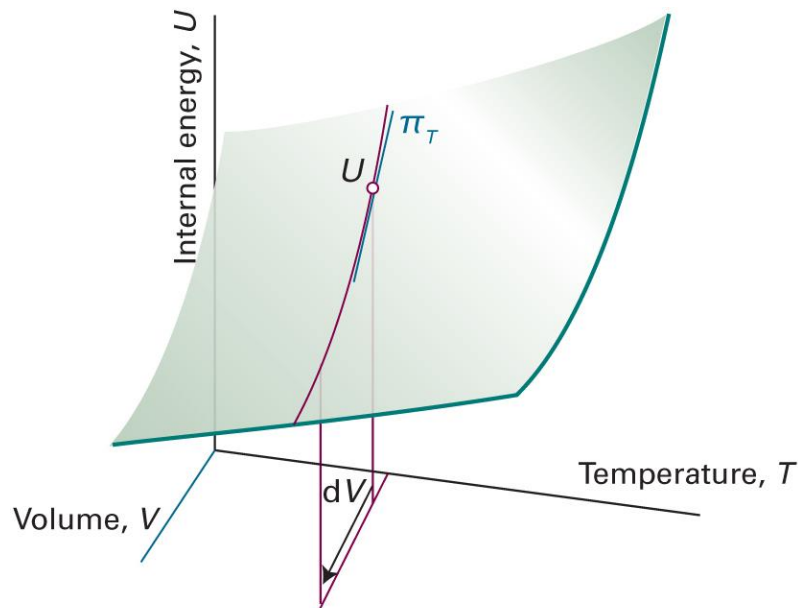
Meaning of $(\partial U / \partial T)_V$



$$dU = C_V dT \text{ (at constant } V\text{)}$$

$$\left(\frac{\partial U}{\partial T} \right)_V = C_V$$

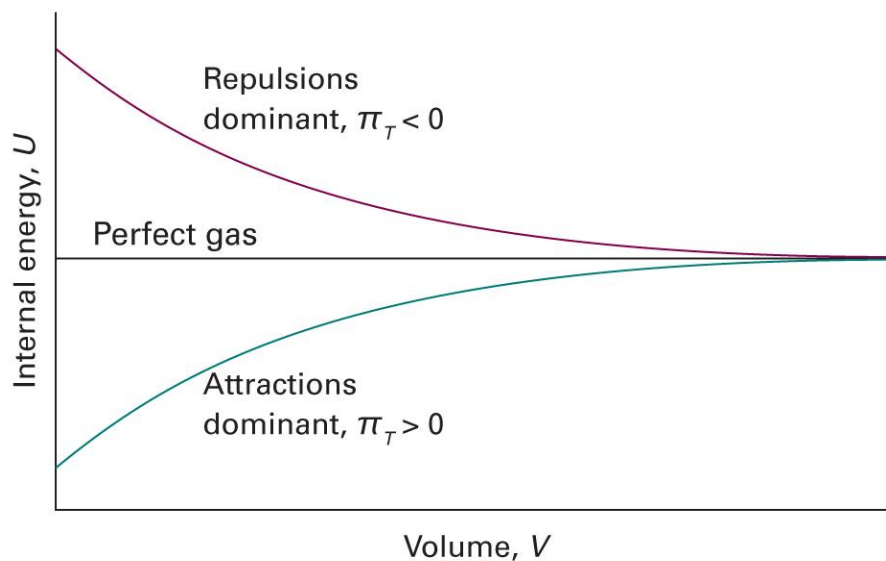
Meaning of $(\partial U / \partial V)_T$



variation of the internal energy of a substance
as its volume is changed at constant temperature

Internal Pressure

$$\pi_T = \left(\frac{\partial U}{\partial V} \right)_T$$



Use of Differentials

$$dU = \underbrace{\left(\frac{\partial U}{\partial V}\right)_T}_{\pi_T} dV + \underbrace{\left(\frac{\partial U}{\partial T}\right)_V}_{C_V} dT$$

$$dU = \pi_T dV + C_V dT$$

At constant pressure,

$$\left(\frac{\partial U}{\partial T}\right)_p = \pi_T \left(\frac{\partial V}{\partial T}\right)_p + C_V$$

Response Functions

- Expansion coefficient

$$\alpha = \frac{1}{V} \left(\frac{\partial V}{\partial T} \right)_p$$

- Isothermal compressibility

$$\kappa_T = -\frac{1}{V} \left(\frac{\partial V}{\partial p} \right)_T$$

Some Data

	$\alpha / (10^{-4} \text{ K}^{-1})$	$\kappa_T / (10^{-6} \text{ bar}^{-1})$
<i>Liquids:</i>		
Benzene	12.4	90.9
Water	2.1	49.0
<i>Solids:</i>		
Diamond	0.030	0.185
Lead	0.861	2.18

General Expression

$$\left(\frac{\partial U}{\partial T}\right)_p = \pi_T \left(\frac{\partial V}{\partial T}\right)_p + C_V$$

Using the definition of the expansion coefficient α ,

$$\left(\frac{\partial U}{\partial T}\right)_p = \alpha \pi_T V + C_V$$

For a perfect gas, $\pi_T = 0$ and

$$\left(\frac{\partial U}{\partial T}\right)_p = C_V = \left(\frac{\partial U}{\partial T}\right)_V$$

Example 2D.2

Derive an expression for the expansion coefficient of a perfect gas.

$$\alpha = \frac{1}{V} \left(\frac{\partial V}{\partial T} \right)_p$$

For a perfect gas, $V = nRT/p$

$$\left(\frac{\partial V}{\partial T} \right)_p = \left(\frac{\partial}{\partial T} \left(\frac{nRT}{p} \right) \right)_p = \frac{nR}{p}$$

Thus,

$$\alpha = \frac{1}{V} \frac{nR}{p} = \frac{nR}{nRT} = \frac{1}{T}$$

Summary for Chapter 3 (1)

First Law of Thermodynamics

$$dU = \delta q + \delta w$$

Work

$$\delta w = -p_{\text{ex}}dV + \delta w_{\text{add}}$$

- Constant- p change
- Reversible change

Heat

- Constant- V change

$$\delta q_V = C_V dT$$

$$C_V = (\partial U / \partial T)_V$$

Thermodynamics

Mathematics

State function

Exact differential

Path function

Inexact differential

Differential form of U

$$dU = \left(\frac{\partial U}{\partial V} \right)_T dV + \left(\frac{\partial U}{\partial T} \right)_V dT$$

- Internal pressure π_T
- Response functions α , κ_T

Enthalpy H

$$H = U + pV$$

- Energy required to create a system (U) and to make a room for the system (pV)
- Since U , p , and V are state functions, H is also a state function
- Molar enthalpy $H_m = H/n$

Gas-Involving Processes

$$H = U + pV$$

For a perfect gas,

$$H = U + nRT$$

that is,

$$\Delta H \approx \Delta U + RT\Delta n_g$$

Infinitesimal Change

For an infinitesimal change in the state,

$$U \rightarrow U + dU$$

$$p \rightarrow p + dp$$

$$V \rightarrow V + dV$$

$$H \rightarrow H + dH$$

From $H = U + pV$,

$$H + dH = (U + dU) + (p + dp)(V + dV)$$

Infinitesimal Change in H

$$H + dH = (U + dU) + (p + dp)(V + dV)$$

$$H + dH = U + dU + pV + pdV + Vdp + \cancel{dpdV}$$

too small

Since $H = U + pV$,

$$dH = dU + pdV + Vdp$$

From the first law of thermodynamics,

$$dH = \delta q + \delta w + pdV + Vdp$$

Reversible Isobaric Process

$$dH = \delta q + \delta w + p dV + V dp$$

For a reversible change,

$$\delta w = -p_{\text{ex}} dV = -p dV$$

so that

$$dH = \delta q - p dV + p dV + V dp = \delta q + V dp$$

At constant pressure ($dp = 0$),

$$dH = \delta q_p$$



Enthalpy Change

$$dH = \delta q_p$$

Integration leads to

$$\Delta H = \int_{\text{state } i}^{\text{state } f} dH = \int_{\text{state } i}^{\text{state } f} \delta q_p = q_p$$

Heat Capacity at Constant p

Since $\delta q_p = C_p dT$,

$$dH = \delta q_p = C_p dT$$

which means

$$C_p = \left(\frac{\partial H}{\partial T} \right)_p$$

The change in enthalpy is

$$\Delta H = \int_{\text{state } i}^{\text{state } f} dH = \int_{\text{state } i}^{\text{state } f} (C_p dT) = \int_{T_i}^{T_f} C_p dT$$

At Constant p

$$\Delta H = \int_{T_i}^{T_f} C_p \, dT$$

If C_p is constant,

$$\Delta H = \int_{T_i}^{T_f} C_p \, dT = C_p \int_{T_i}^{T_f} dT = C_p (T_f - T_i) = C_p \Delta T$$

Since $\Delta H = q_p$ at constant pressure,

$$q_p = C_p \Delta T$$

T -Dependence of $C_{p,m}$

Empirical expression:

$$C_{p,m} = a + bT + \frac{c}{T^2}$$

	$a / (\text{J K}^{-1} \text{mol}^{-1})$	$b / (10^{-3} \text{J K}^{-2} \text{mol}^{-1})$	$c / (10^5 \text{J K mol}^{-1})$
C(s, graphite)	16.86	4.77	-8.54
CO ₂ (g)	44.22	8.79	-8.62
H ₂ O(l)	75.29	0	0
N ₂ (g)	28.58	3.77	-0.50

Example 2B.2

What is the change in molar enthalpy of N_2 when it is heated from 25°C to 100°C ? Use the heat capacity information in the previous table.

$$\Delta H = \int_{T_i}^{T_f} C_p \, dT$$

$$C_{p,m} = a + bT + \frac{c}{T^2}$$

Hence,

$$\Delta H_m = \int_{T_i}^{T_f} C_{p,m} \, dT = \int_{T_i}^{T_f} \left(a + bT + \frac{c}{T^2} \right) dT = \left[aT + \frac{b}{2}T^2 - \frac{c}{T} \right]_{T_i}^{T_f}$$

Example 2B.2

$$\Delta H_m = \left(aT_f + \frac{b}{2}T_f^2 - \frac{c}{T_f} \right) - \left(aT_i + \frac{b}{2}T_i^2 - \frac{c}{T_i} \right)$$

$$T_i = 25^\circ\text{C} = 298\text{ K}, T_f = 100^\circ\text{C} = 373\text{ K}$$

Substitution leads to

$$\Delta H_m = H_m(373\text{ K}) - H_m(298\text{ K}) = 2.20\text{ kJ mol}^{-1}$$

Changes in Enthalpy

For $H = H(p, T)$,

$$dH = \underbrace{\left(\frac{\partial H}{\partial p}\right)_T}_{?} dp + \underbrace{\left(\frac{\partial H}{\partial T}\right)_p}_{C_p} dT$$

Let $dH = 0$ (constant enthalpy).

$$\left(\frac{\partial H}{\partial p}\right)_T dp = -C_p dT$$

Constant Enthalpy

At constant enthalpy,

$$\left(\frac{\partial H}{\partial p}\right)_T dp = -C_p dT$$

$$\left(\frac{\partial H}{\partial p}\right)_T = -C_p \left(\frac{\partial T}{\partial p}\right)_H$$

Define the **Joule-Thomson coefficient**: $\mu = (\partial T / \partial p)_H$

$$\left(\frac{\partial H}{\partial p}\right)_T = -C_p \mu$$

Changes in Enthalpy

For $H = H(p, T)$,

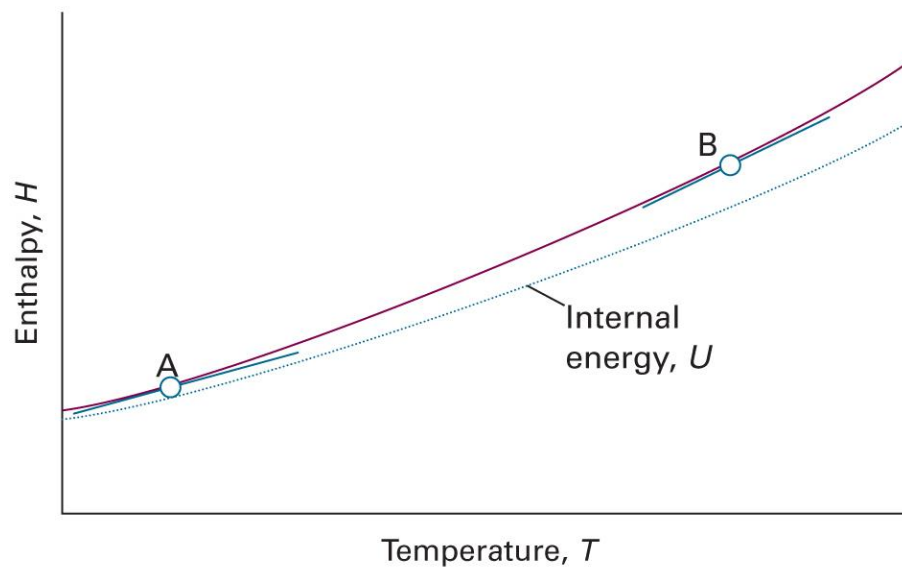
$$dH = \underbrace{\left(\frac{\partial H}{\partial p}\right)_T}_{-C_p\mu} dp + \underbrace{\left(\frac{\partial H}{\partial T}\right)_p}_{C_p} dT$$

$$dH = -C_p\mu dp + C_p dT$$

(Compare: $dU = \pi_T dV + C_V dT$)

C_V and C_p

$$C_V = \left(\frac{\partial U}{\partial T} \right)_V, \quad C_p = \left(\frac{\partial H}{\partial T} \right)_p$$



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 2B.3)

Case of a Perfect Gas

$$C_p - C_V = \left(\frac{\partial H}{\partial T} \right)_p - \left(\frac{\partial U}{\partial T} \right)_p$$

For a perfect gas,

$$H = U + pV = U + nRT$$

so that

$$\begin{aligned} C_p - C_V &= \left(\frac{\partial (U + nRT)}{\partial T} \right)_p - \left(\frac{\partial U}{\partial T} \right)_p \\ &= \left(\frac{\partial U}{\partial T} \right)_p + nR - \left(\frac{\partial U}{\partial T} \right)_p = nR \end{aligned}$$